

JMA2402	APPLIED LINEAR ALGEBRA (For CSE, CSE (AI&ML), CSE (CS), CS&BS, AI&DS, IT & ECE)	L	T	P	C
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COURSE OBJECTIVES:

- To impart knowledge of consistency of system of equations through Rouche-Capelli theorem.
- To familiarize students in the concepts of vector spaces, bases and dimension.
- To enable students understand concepts of linear transformation, Eigenvalues, Eigenvectors and Diagonalizability.
- To introduce inner product spaces, orthogonalization and least square approximations.

UNIT I	MATRICES AND SYSTEM OF LINEAR EQUATIONS	9+3
Matrices – Row Echelon form – Rank – Non-homogeneous system of linear equations – Homogeneous system of linear equations – Rouche-Capelli theorem (without proof) – Consistency – Gauss Jordan method – Gauss Seidel method		
UNIT II	VECTOR SPACES	9+3
Definition of field – Vector spaces – Subspaces – Properties – Linear combinations – Linear independence and linear dependence – Span – Bases and dimensions		
UNIT III	LINEAR TRANSFORMATION	9+3
Linear transformation – Properties – Kernel and Range – One-to-One correspondence – Rank and Nullity – Dimension theorem		
UNIT IV	DIAGONALIZATION	9+3
Matrix representations of linear transformations – Eigenvalues and eigenvectors – Diagonalizability		

UNIT V INNER PRODUCT SPACES**9+3**

Inner products and norms – Gram Schmidt orthogonalization process – Least square approximation

TOTAL: 60 PERIODS**COURSE OUTCOMES:**

At the end of the course, the student should be able to:

- Apply Rouché-Capelli theorem for solving system of equations.
- Solve problems occurring in vector spaces and subspaces.
- Conceptualize linear transformation and use it in real-time problems.
- Demonstrate efficient use of matrix representations and diagonalization.
- Solve problems using orthogonalization process and use least square approximations.

TEXT BOOKS:

1. Grewal B.S., “Higher Engineering Mathematics”, Khanna Publishers, New Delhi, 44th Edition, 2021.
2. Kreyszig Erwin., "Advanced Engineering Mathematics", John Wiley and Sons, 10th Edition, New Delhi, 2016.
3. Friedberg. A.H., Insel. A.J. and Spence. L., “Linear Algebra”, Prentice Hall of India, New Delhi, 2004.

REFERENCES:

1. Kumaresan. S., “Linear Algebra – A Geometric Approach”, Prentice – Hall of India, New Delhi, Reprint, 2010.
2. 4. Lay D.C., “Linear Algebra and its Applications”, 5th Edition, Pearson Education, 2015.

WEB REFERENCES:

- <https://nptel.ac.in/courses/111107105/>
- <https://nptel.ac.in/courses/108104174/>
- <https://nptel.ac.in/courses/111104137/>
- <https://nptel.ac.in/courses/111106135/>
- <https://nptel.ac.in/courses/111108157/>

Mapping of CO with PO

	Program Outcomes											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	-	-	-	2	-	-	1
CO2	3	2	2	2	3	-	-	-	2	-	-	1
CO3	3	2	2	2	3	-	-	-	2	-	-	1
CO4	3	2	2	2	3	-	-	-	2	-	-	1
CO5	3	2	2	2	3	-	-	-	2	-	-	1
AVG	3	2	2	2	3	-	-	-	2	-	-	1